

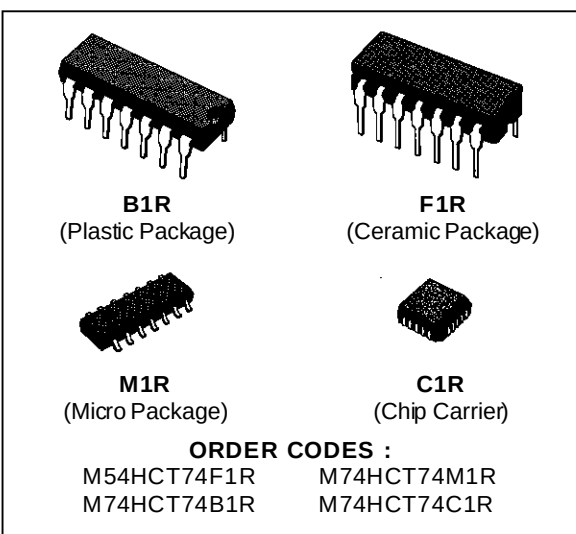
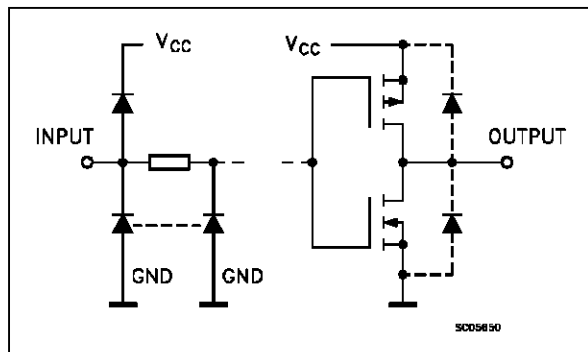
DUAL D TYPE FLIP FLOP WITH PRESET AND CLEAR

- HIGH SPEED
 $f_{MAX} = 53 \text{ MHz (TYP.) AT } V_{CC} = 5 \text{ V}$
- LOW POWER DISSIPATION
 $I_{CC} = 2 \mu\text{A (MAX.) AT } T_A = 25^\circ\text{C}$
- COMPATIBLE WITH TTL OUTPUTS
 $V_{IH} = 2 \text{ V (MIN.) } V_{IL} = 0.8 \text{ V (MAX)}$
- OUTPUT DRIVE CAPABILITY
10 LSTTL LOADS
- SYMMETRICAL OUTPUT IMPEDANCE
 $|I_{OH}| = I_{OL} = 4 \text{ mA (MIN.)}$
- BALANCED PROPAGATION DELAYS
 $t_{PLH} = t_{PHL}$
- PIN AND FUNCTION COMPATIBLE
WITH 54/74LS74

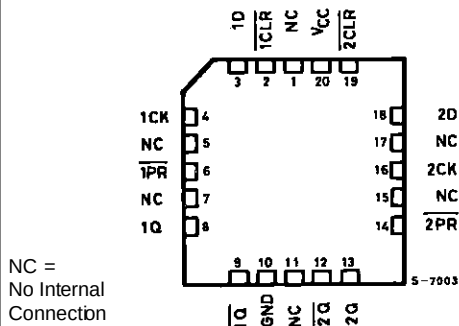
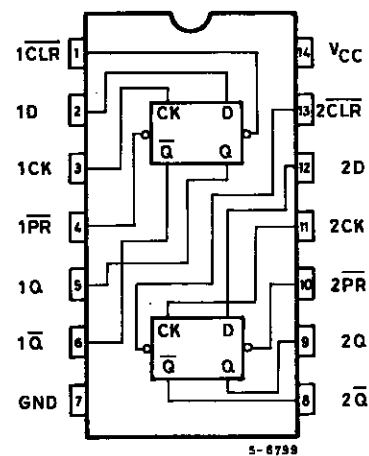
DESCRIPTION

The M54/74HCT74 is a high speed CMOS DUAL D TYPE FLOP WITH PRESET AND CLEAR fabricated in silicon gate C²MOS technology. It has the same high speed performance of LSTTL combined with true CMOS low power consumption. A signal on the D INPUT is transferred to the Q OUTPUT during the positive going transition of the clock pulse. CLEAR and PRESET are independent of the clock and accomplished by a low on the appropriate input. All inputs are equipped with protection circuits against static discharge and transient excess voltage. This integrated circuit has input and output characteristics that are fully compatible with 54/74 LSTTL logic families. M54/74HCT devices are designed to directly interface HSC²MOS systems with TTL and NMOS components. They are also plug in replacements for LSTTL devices giving a reduction of power consumption.

INPUT AND OUTPUT EQUIVALENT CIRCUIT



PIN CONNECTIONS (top view)



TRUTH TABLE

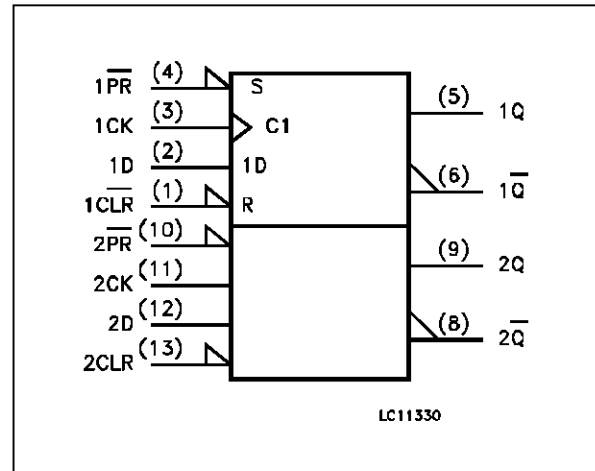
INPUTS				OUTPUTS		FUNCTION
CLR	PR	D	CK	Q	$\bar{Q}$	
L	H	X	X	L	H	CLEAR
H	L	X	X	H	L	PRESET
L	L	X	X	H	H	
H	H	L	$\downarrow$	L	H	
H	H	H	$\downarrow$	H	L	
H	H	X	$\downarrow$	Q_n	$\bar{Q}_n$	NO CHANGE

X: Don't Care

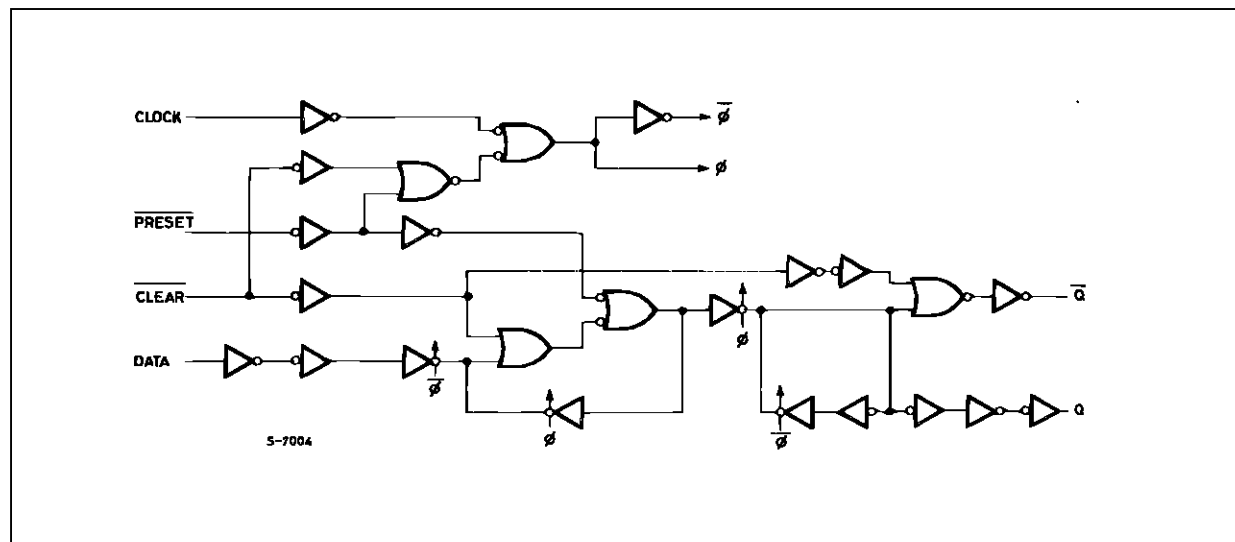
PIN DESCRIPTION

PIN No	SYMBOL	NAME AND FUNCTION
1, 13	1CLR, 2CLR	Asynchronous Reset - Direct Input
2, 12	1D, 2D	Data Inputs
3, 11	1CK, 2CK	Clock Input (LOW-to-HIGH, Edge-Triggered)
4, 10	1PR, 2PR	Asynchronous Set - Direct Input
5, 9	1Q, 2Q	True Flip-Flop Output
6, 8	1 $\bar{Q}$, 2 $\bar{Q}$	Complement Flip-Flop Outputs
7	GND	Ground (0V)
14	V _{CC}	Positive Supply Voltage

IEC LOGIC SYMBOL



LOGIC DIAGRAM



ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V_{CC}	Supply Voltage	-0.5 to +7	V
V_I	DC Input Voltage	-0.5 to $V_{CC} + 0.5$	V
V_O	DC Output Voltage	-0.5 to $V_{CC} + 0.5$	V
I_{IK}	DC Input Diode Current	± 20	mA
I_{OK}	DC Output Diode Current	± 20	mA
I_O	DC Output Source Sink Current Per Output Pin	± 25	mA
I_{CC} or I_{GND}	DC V_{CC} or Ground Current	± 50	mA
P_D	Power Dissipation	500 (*)	mW
T_{stg}	Storage Temperature	-65 to +150	°C
T_L	Lead Temperature (10 sec)	300	°C

Absolute Maximum Ratings are those values beyond which damage to the device may occur. Functional operation under these condition is not implied.

(*) 500 mW: $\equiv 65^\circ\text{C}$ derate to 300 mW by 10mW/°C: 65°C to 85°C

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter	Value	Unit
V_{CC}	Supply Voltage	4.5 to 5.5	V
V_I	Input Voltage	0 to V_{CC}	V
V_O	Output Voltage	0 to V_{CC}	V
T_{op}	Operating Temperature: M54HC Series	-55 to +125	°C
	M74HC Series	-40 to +85	°C
t_r, t_f	Input Rise and Fall Time ($V_{CC} = 4.5$ to 5.5V)	0 to 500	ns

DC SPECIFICATIONS

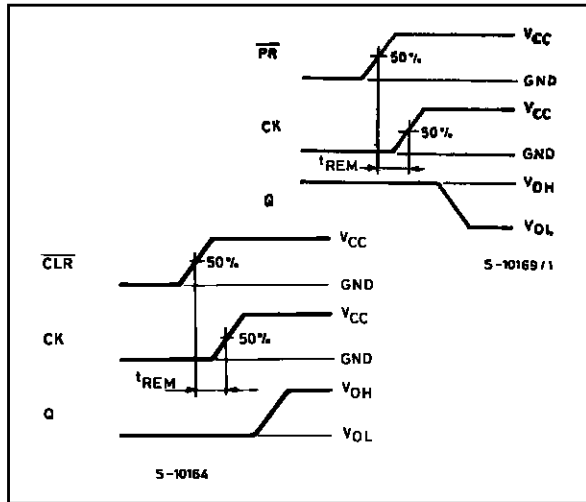
Symbol	Parameter	Test Conditions				Value					Unit	
		V _{CC} (V)			T _A = 25 °C 54HC and 74HC			-40 to 85 °C 74HC		-55 to 125 °C 54HC		
					Min.	Typ.	Max.	Min.	Max.	Min.		Max.
V _{IH}	High Level Input Voltage	4.5 to 5.5			2.0			2.0		2.0		V
V _{IL}	Low Level Input Voltage	4.5 to 5.5					0.8		0.8		0.8	V
V _{OH}	High Level Output Voltage	4.5	V _I = V _{IH} or V _{IL}	I _O =-20 μA	4.4	4.5		4.4		4.4		V
				I _O =-4.0 mA	4.18	4.31		4.13		4.10		
V _{OL}	Low Level Output Voltage	4.5	V _I = V _{IH} or V _{IL}	I _O = 20 μA		0.0	0.1		0.1		0.1	V
				I _O = 4.0 mA		0.17	0.26		0.33		0.4	
I _I	Input Leakage Current	5.5	V _I = V _{CC} or GND				±0.1		±1		±1	μA
I _{CC}	Quiescent Supply Current	5.5	V _I = V _{CC} or GND				2		20		40	μA
ΔI _{CC}	Additional worst case supply current	5.5	Per Input pin V _I = 0.5V or V _I = 2.4V Other Inputs at V _{CC} or GND I _O = 0				2.0		2.9		3.0	mA

AC ELECTRICAL CHARACTERISTICS ($C_L = 50 \text{ pF}$, Input $t_r = t_f = 6 \text{ ns}$)

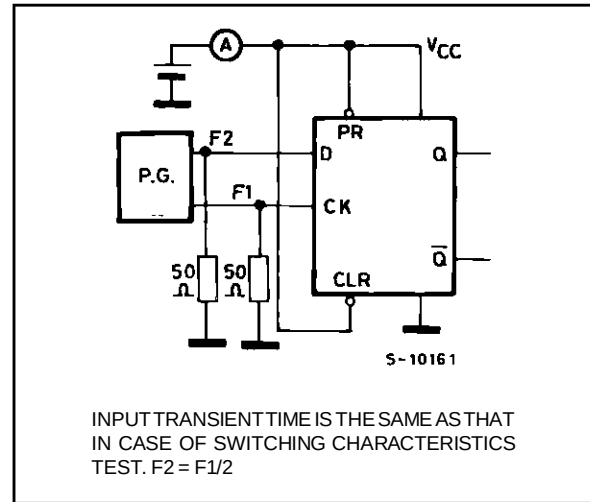
Symbol	Parameter	Test Conditions		Value							Unit
		V _{CC} (V)		T _A = 25 °C 54HC and 74HC			-40 to 85 °C 74HC		-55 to 125 °C 54HC		
				Min.	Typ.	Max.	Min.	Max.	Min.	Max.	
t _{TLH} t _{THL}	Output Transition Time	4.5			8	15		19		22	ns
t _{PLH} t _{PHL}	Propagation Delay Time (CLOCK - Q)	4.5			21	33		41		50	ns
t _{PLH} t _{PHL}	Propagation Delay Time ($\overline{\text{CL}}$, $\overline{\text{PR}}$ - Q, $\overline{\text{Q}}$)	4.5			18	30		38		45	ns
f _{MAX}	Maximum Clock Frequency	4.5		27	48		22		18		MHz
t _{W(H)} t _{W(L)}	Minimum Pulse Width (CLOCK)	4.5			6	15		19		23	ns
t _{W(L)}	Minimum Pulse Width ($\overline{\text{CL}}$, $\overline{\text{PR}}$)	4.5			8	15		19		23	ns
t _s	Minimum Set-up Time	4.5			7	15		19		23	ns
t _h	Minimum Hold Time	4.5				0		0		0	ns
t _{REM}	Minimum Removal Time ($\overline{\text{CL}}$, $\overline{\text{PR}}$)	4.5			1	5	5	6	5	8	ns
C _{IN}	Input Capacitance				5	10		10		10	pF
C _{PD} (*)	Power Dissipation Capacitance				32						pF

(*) C_{PD} is defined as the value of the IC's internal equivalent capacitance which is calculated from the operating current consumption without load. (Refer to Test Circuit). Average operating current can be obtained by the following equation. $I_{CC(opr)} = C_{PD} \cdot V_{CC} \cdot f_{IN} + I_{CC}/2$ (per FLIP/FLOP)

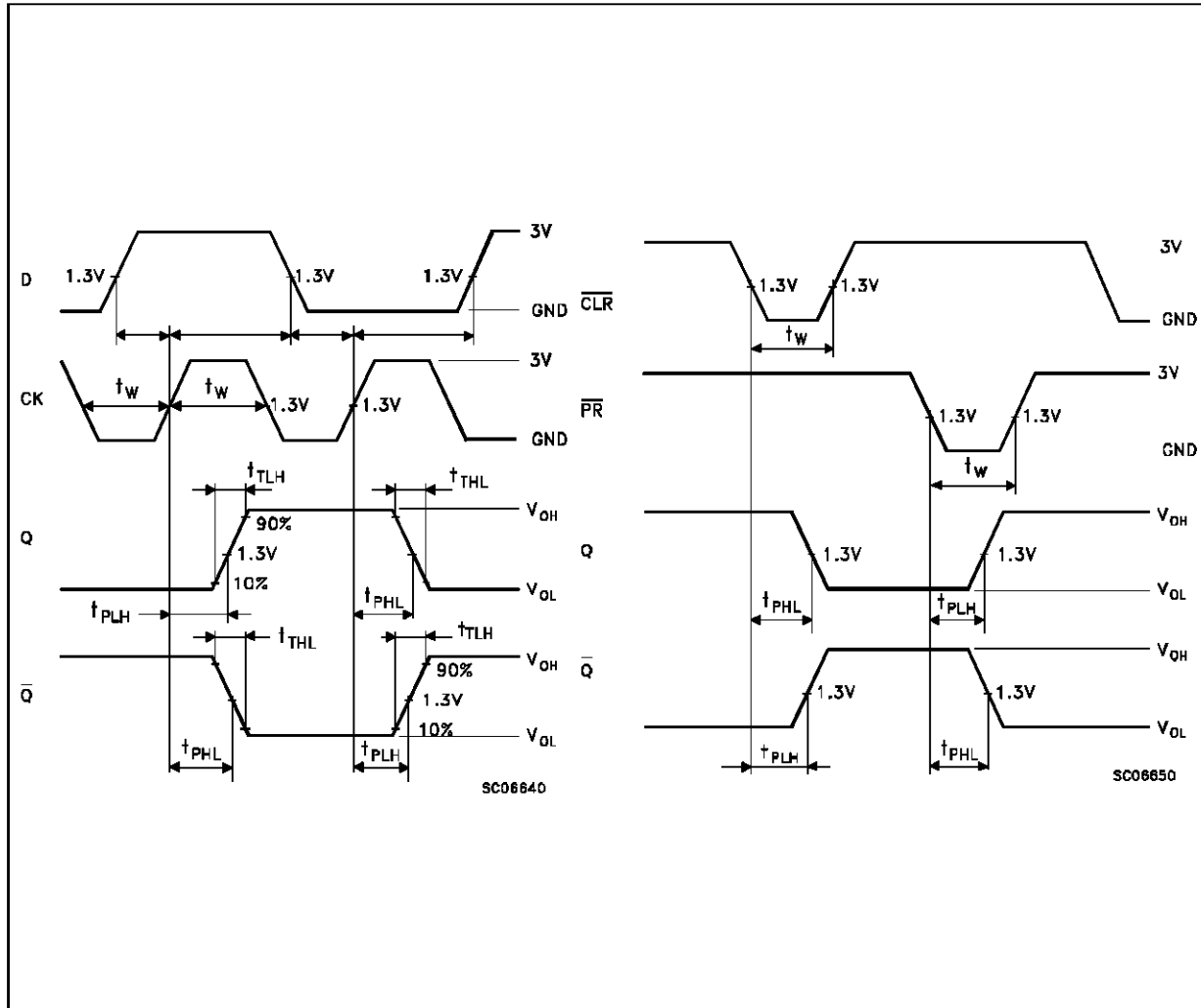
SWITCHING CHARACTERISTICS TEST WAVEFORM



TEST CIRCUIT I_{CC} (Opr.)

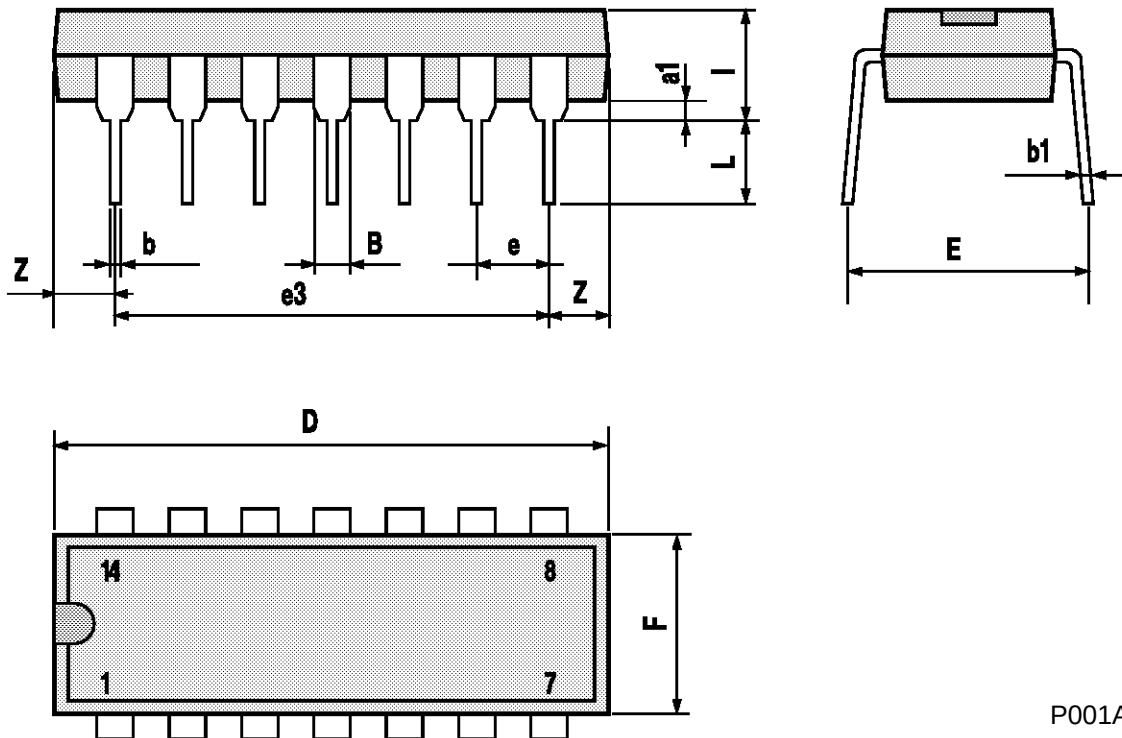


SWITCHING CHARACTERISTICS TEST WAVEFORM



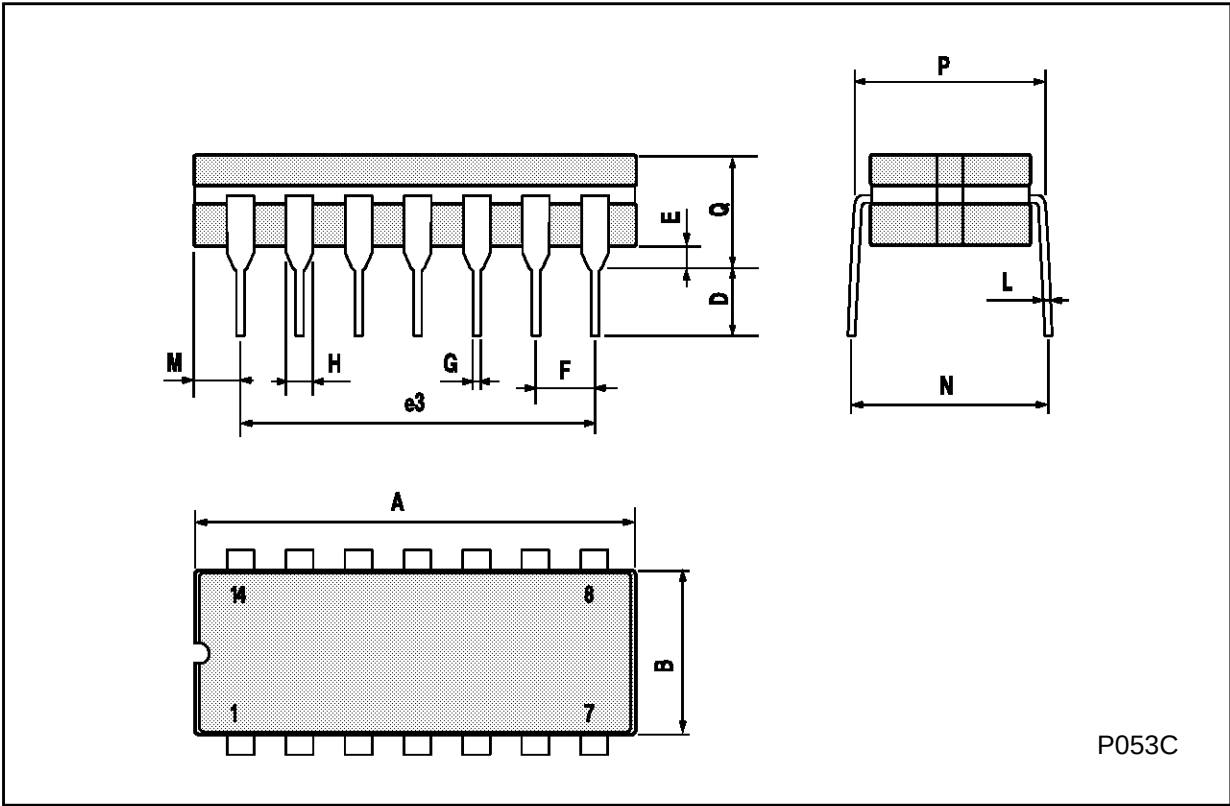
Plastic DIP14 MECHANICAL DATA

DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
a1	0.51			0.020		
B	1.39		1.65	0.055		0.065
b		0.5			0.020	
b1		0.25			0.010	
D			20			0.787
E		8.5			0.335	
e		2.54			0.100	
e3		15.24			0.600	
F			7.1			0.280
I			5.1			0.201
L		3.3			0.130	
Z	1.27		2.54	0.050		0.100



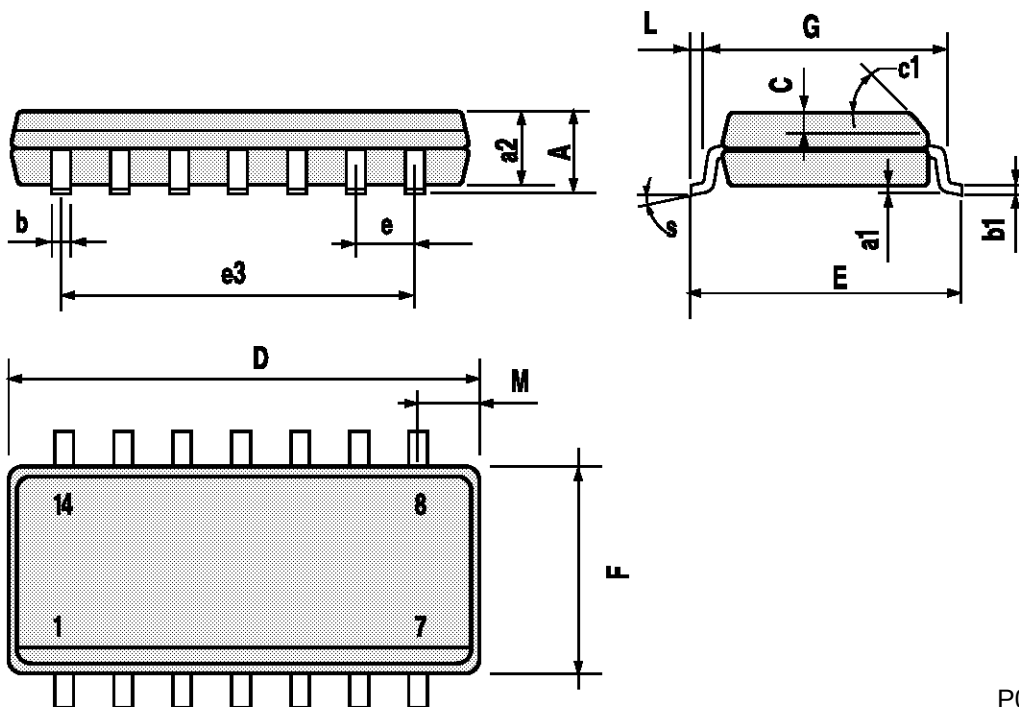
Ceramic DIP14/1 MECHANICAL DATA

DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A			20			0.787
B			7.0			0.276
D		3.3			0.130	
E	0.38			0.015		
e3		15.24			0.600	
F	2.29		2.79	0.090		0.110
G	0.4		0.55	0.016		0.022
H	1.17		1.52	0.046		0.060
L	0.22		0.31	0.009		0.012
M	1.52		2.54	0.060		0.100
N			10.3			0.406
P	7.8		8.05	0.307		0.317
Q			5.08			0.200



SO14 MECHANICAL DATA

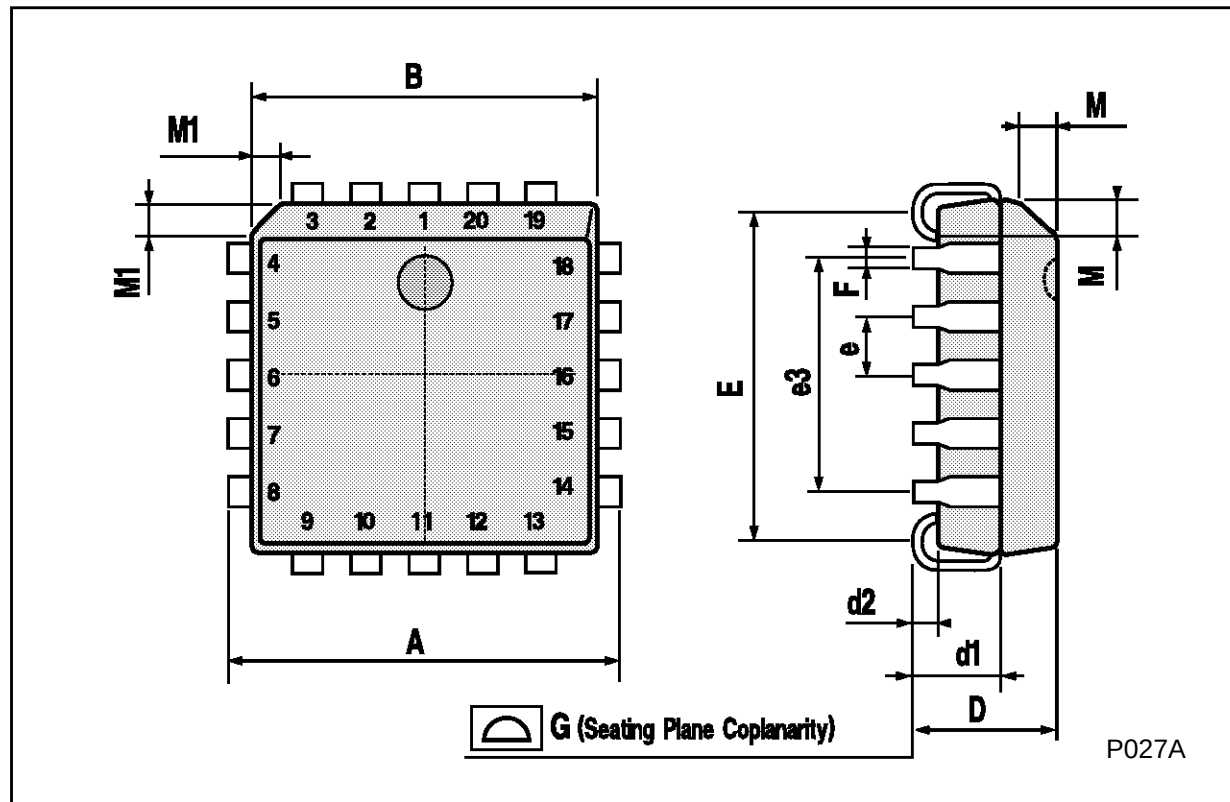
DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A			1.75			0.068
a1	0.1		0.2	0.003		0.007
a2			1.65			0.064
b	0.35		0.46	0.013		0.018
b1	0.19		0.25	0.007		0.010
C		0.5			0.019	
c1	45° (typ.)					
D	8.55		8.75	0.336		0.344
E	5.8		6.2	0.228		0.244
e		1.27			0.050	
e3		7.62			0.300	
F	3.8		4.0	0.149		0.157
G	4.6		5.3	0.181		0.208
L	0.5		1.27	0.019		0.050
M			0.68			0.026
S	8° (max.)					



P013G

PLCC20 MECHANICAL DATA

DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A	9.78		10.03	0.385		0.395
B	8.89		9.04	0.350		0.356
D	4.2		4.57	0.165		0.180
d1		2.54			0.100	
d2		0.56			0.022	
E	7.37		8.38	0.290		0.330
e		1.27			0.050	
e3		5.08			0.200	
F		0.38			0.015	
G			0.101			0.004
M		1.27			0.050	
M1		1.14			0.045	



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